

Recap

$$H = E[I(x)] = \sum_x p(x) I(x) = - \sum_x p(x) \log p(x)$$

- Entropy

$$H(x, y) = - \sum_{x, y} p(x, y) \cdot \log p(x, y)$$

- Joint Entropy, Conditional Entropy

$$H(Y|X) = \sum_x p(x) H(Y|X=x)$$

an informative feature leads to high $I(X_i, Y)$

average conditional entropy classification

- Mutual Information

$$I(X_i, Y) = H(Y) - H(Y|X_i)$$

reduction in uncertainty of Y given we observed feature X_i
 X_i has a lot of information (weight in cat/dog)

- Cross Entropy

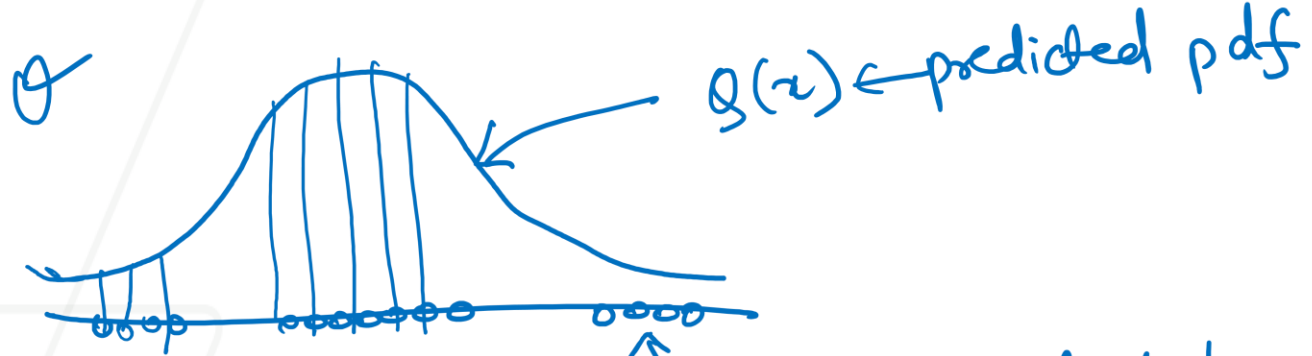
low inefficiency
we are encoding samples from $P(x)$ by actually following $Q(x)$

$$H(Y|X_i) \downarrow \quad H(Y) - H(Y|X_i) \uparrow$$

Optimization

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$$E[a+b+c] = E[a] + E[b] + E[c]$$

$$E[g(x)] = \sum_x g(x) \cdot p(x)$$

$$L(\theta|x) = \prod_{i=1}^n f(x^{(i)}|\theta)$$

maximize

$$\log L(\theta|x) = \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

minimize

$$l(\theta|x) = - \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

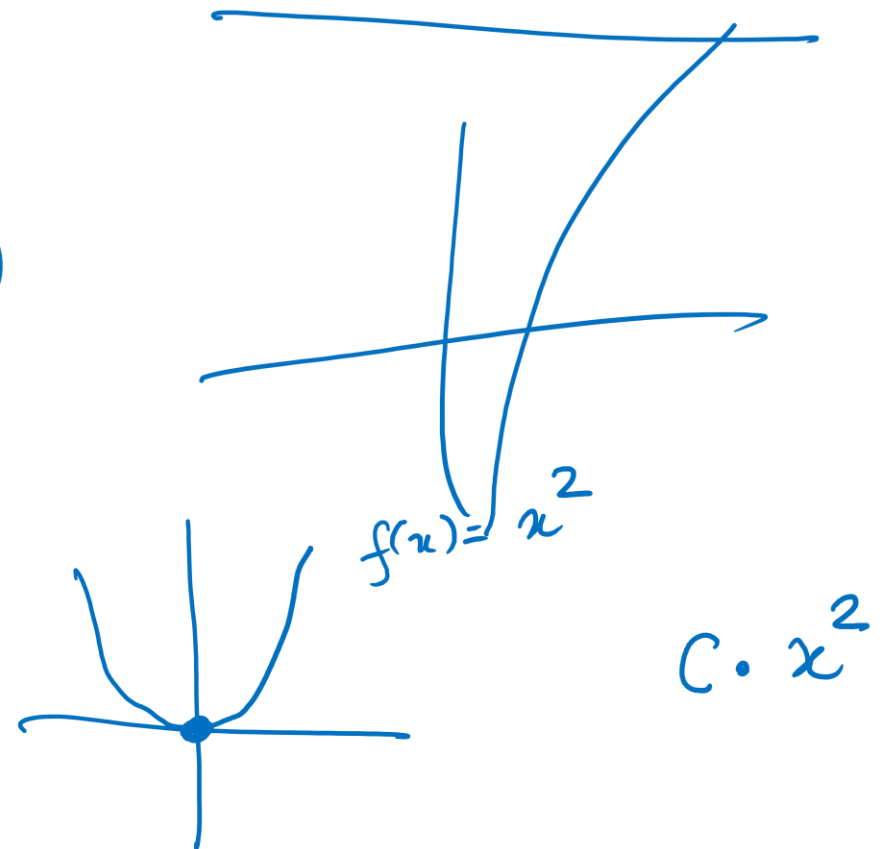
$$= - \frac{1}{n} \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

$$= - E[\log Q(x)]$$

$$= - E[\log P(x) - \log P(x) + \log Q(x)]$$

$$= - E[\log P(x)] - E\left[\log \frac{Q(x)}{P(x)}\right]$$

$$H(P) = - \sum_x p(x) \log P(x) = - \sum_x p(x) \log \frac{Q(x)}{P(x)}$$



①

$$\min l(\theta|x) = \underbrace{-\sum_x p(x) \log p(x)}_{H[p]} - \underbrace{\sum_x p(x) \log \frac{q(x)}{p(x)}}_{KL[p][q]}$$

likelihood minimization = Min. $H[p] + KL[p][q]$

Minimizing CE is the same as minimizing -ve log likelihood.

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = H(P) + KL[P][Q]$$

Handwritten annotations:

- $H(p, q)$ is circled and labeled "loss function" and "CE" (Cross Entropy).
- The sum $\sum_{x \in \mathcal{X}} p(x) \log q(x)$ is boxed and labeled "objective function".
- The term $KL[P][Q]$ is bracketed and labeled "non negative".
- A handwritten expression $\sum_x p(x) \log \frac{1}{1/q(x)}$ is shown with an arrow pointing to the $\log q(x)$ term in the boxed sum.

This is because:

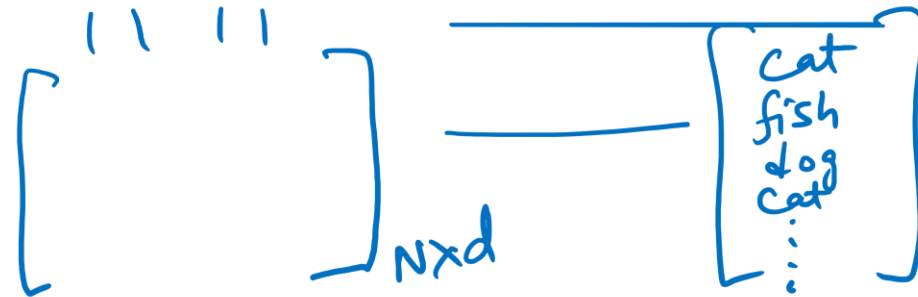
$$H(p, q) = \mathbb{E}_p[l_i] = \mathbb{E}_p \left[\log \frac{1}{q(x_i)} \right]$$

$$H(p, q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

$$H(p, q) = - \sum_x p(x) \log q(x).$$

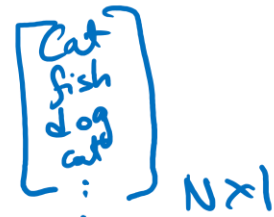
Labeling target values: Label encoding (ordinal) and One-hot encoding

• Input Matrix (X):



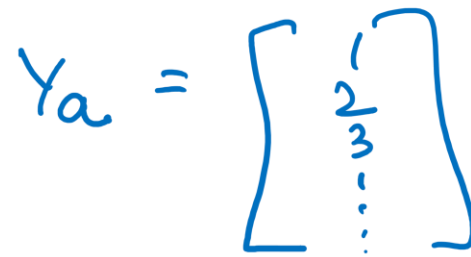
Multiclass classification

• Target Labels (Y):

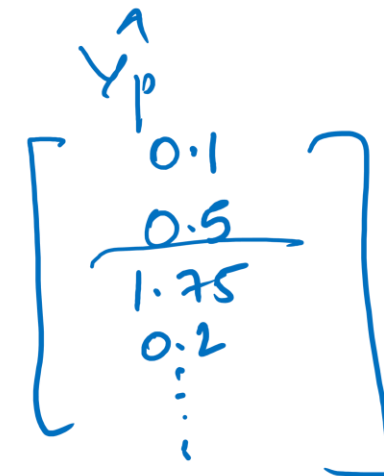


• Label Encoding (Ordinal):

cat $\rightarrow$ 1, fish $\rightarrow$ 2, dog $\rightarrow$ 3



model

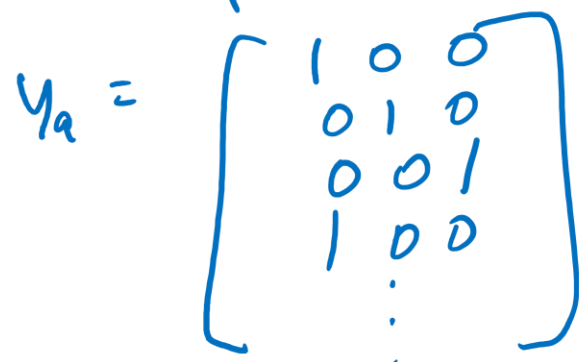


tshirt sizes

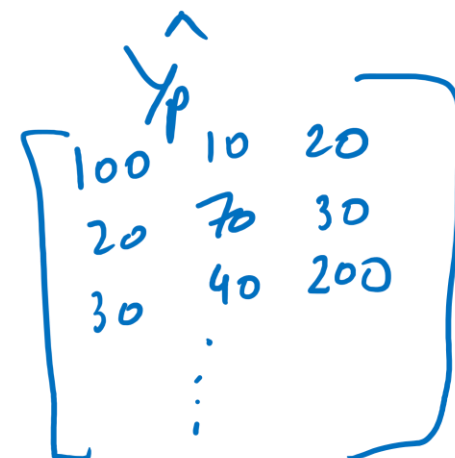
• One-Hot Encoding:

Cat $\rightarrow$ [1 0 0], fish $\rightarrow$ [0 1 0], dog $\rightarrow$ [0 0 1]

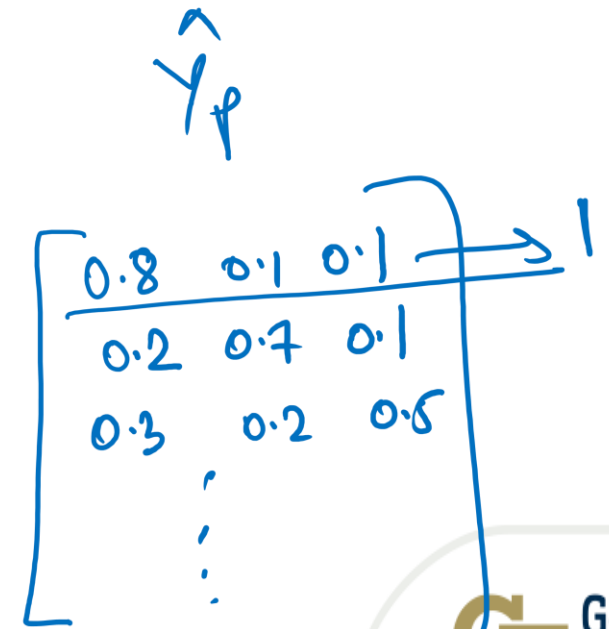
sums up to 1
pm f

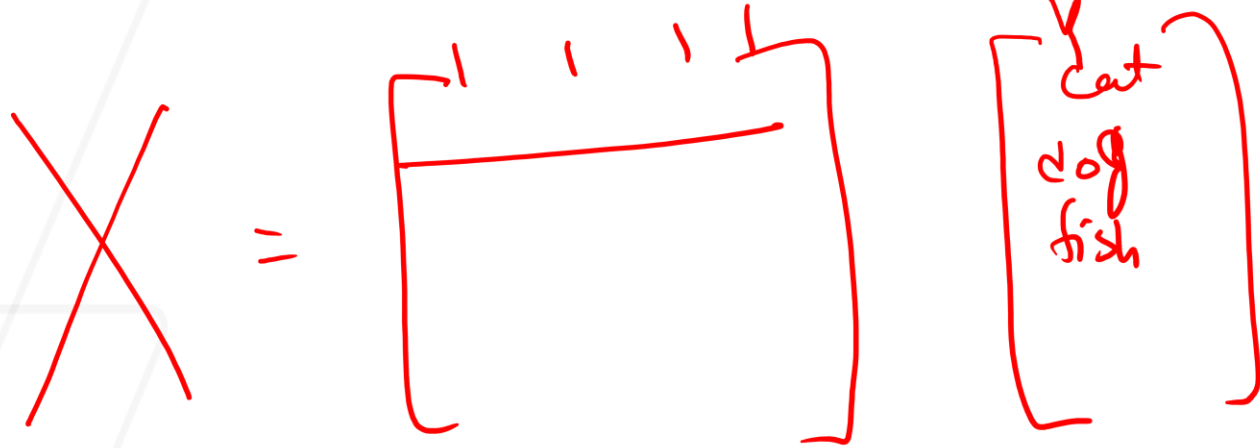


model

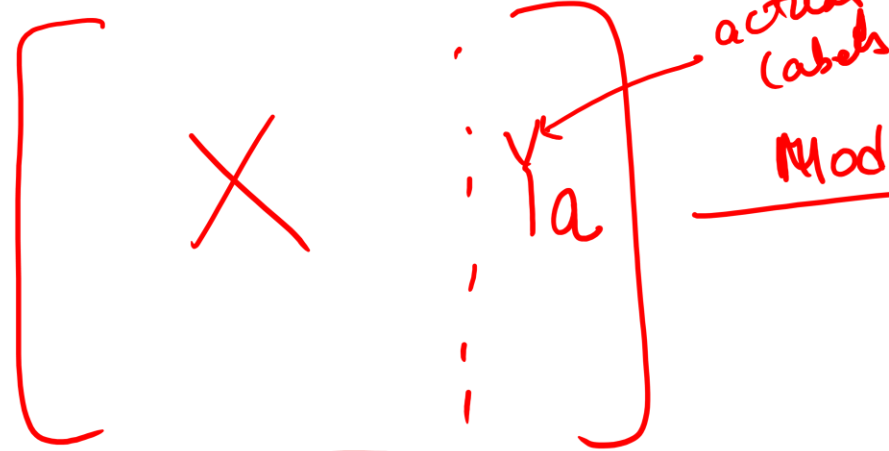


Softmax



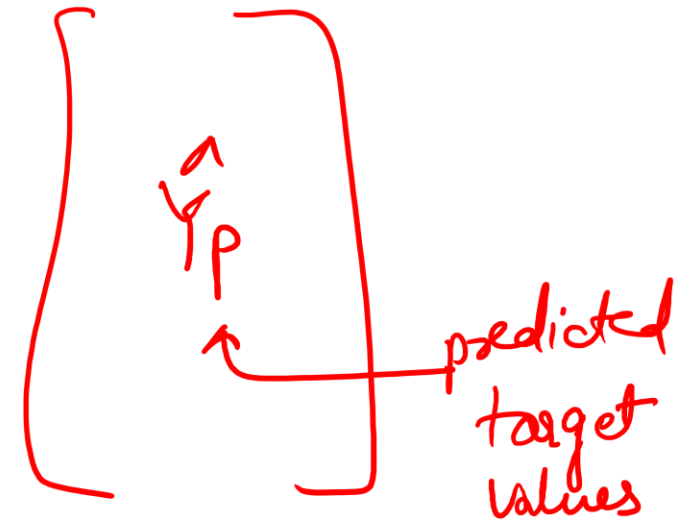


Training data



Model

use first set of parameters that it assumes



How Y_a and $\hat{Y}_p$ vary

as close as possible

Trained Model $\rightarrow$ a new datapoint $\rightarrow \hat{Y}_p$

$f(x)$
Objective
maximize or minimize

Label encoding (One-hot encoding) Loss Computation

Minimize Objective Function:

$$\min \sum_{i=1}^n \| y_a^{(i)} - \hat{y}_p^{(i)T} \|_2^2$$

MSE

1st data point

$$y_a = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\hat{y}_p = \begin{bmatrix} 0.8 & 0.1 & 0.1 \end{bmatrix}$$

$$(1 - 0.8)^2 + (0 - 0.1)^2 + (0 - 0.1)^2 = (0.2)^2$$

In a perfect world: $\hat{y}_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$

$$0 + 0 + 0 + \dots + n = 0$$

Maximize Objective Function:

$$\max \sum_{i=1}^n y_a^{(i)} \cdot \hat{y}_p^{(i)T}$$

$$\min - \sum_{i=1}^n y_a^{(i)} \cdot \hat{y}_p^{(i)T}$$

$$0.8 \times 1 + 0 \times 0.1 + 0 \times 0.1 = 0.8$$

In a perfect world = 1

$$1 + 1 + 1 + \dots + n = n$$

Interpretation:

- Minimum squared Error works but isn't ideal for classification.
- **Minimize negative sum of dot product is also good but Cross-Entropy / NLL is better.**

Why Cross Entropy is better than dot product

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x)$$

$\uparrow$ $y_a^{\{i\}}$ $\uparrow$ $\hat{y}_p^{\{i\}}$

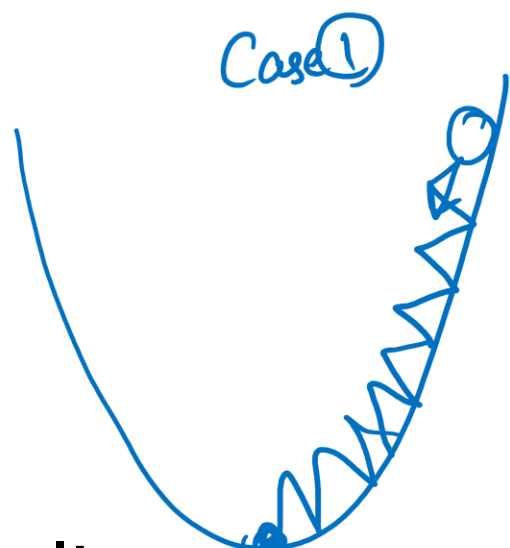
$= \text{CE} = - \sum_{i=1}^n y_a^{\{i\}} \cdot \log \hat{y}_p^{\{i\}T} \text{ --- Case ②}$

$\text{---ve log likelihood}$

$\min - \sum_{i=1}^n y_a^{\{i\}} \cdot \hat{y}_p^{\{i\}T} \text{ --- Case ①}$



Log really really penalizes our objective function



Convergence is quick

Logarithmic penalty encourages big update leading to faster convergence

Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\begin{aligned}\mathbf{KL}[P(S)||Q(S)] &= \sum_s P(s) \log \frac{P(s)}{Q(s)} \\ &= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P, Q) - H(P)\end{aligned}$$

Excess cost in bits paid by encoding according to Q instead of P .

KL Divergence is
a **KIND OF**
distance
measurement

$$-\mathbf{KL}[P||Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is
concave or
convex?

$$\begin{aligned}\sum_s P(s) \log \frac{Q(s)}{P(s)} &\leq \log \sum_s P(s) \frac{Q(s)}{P(s)} \\ &= \log \sum_s Q(s) = \log 1 = 0\end{aligned}$$

By Jensen Inequality

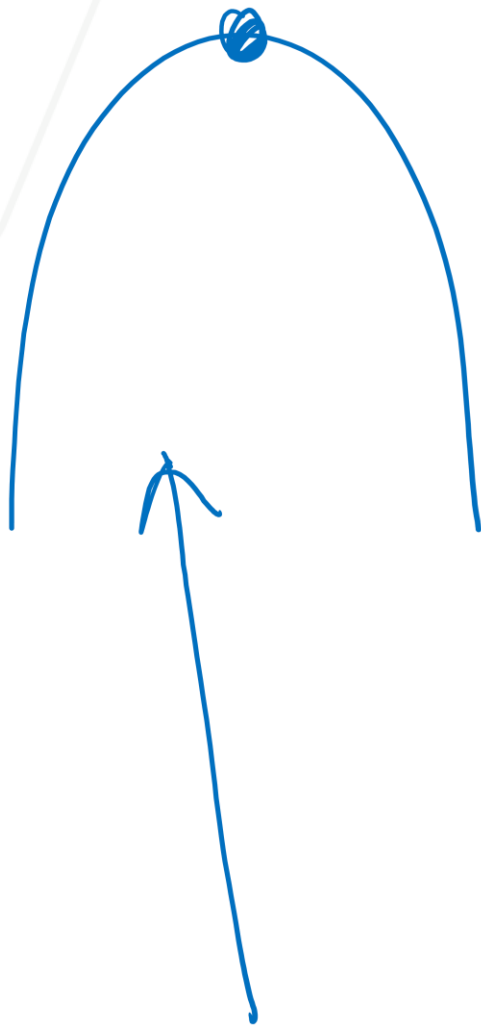
So $\mathbf{KL}[P||Q] \geq 0$. Equality iff $P = Q$

When $P = Q$, $KL[P||Q] = 0$

Concave Function: Jensen Inequality

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$$

$$f(x) = -x^2$$
$$f''(x) < 0 \rightarrow \text{concave}$$



Strictly concave functions have global maximum

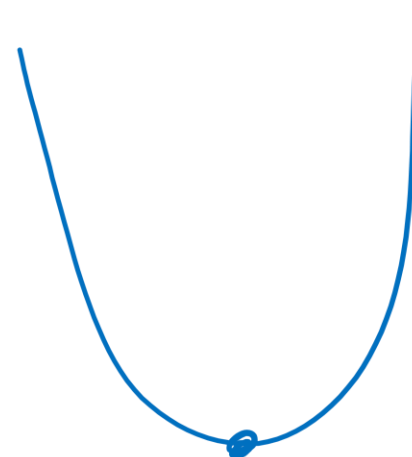
Convex Function: Jensen Inequality

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

$$f(x) = x^2$$

$$x = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$p(x) = \begin{bmatrix} 1/2 & 1/2 \end{bmatrix}$$



$$\begin{aligned} \mathbb{E}[f(x)] &= \sum_x f(x) \cdot p(x) = f(x=1)p(x=1) + f(x=2)p(x=2) \\ &= 1 \cdot 1/2 + 4 \cdot 1/2 = 2.5 \end{aligned}$$

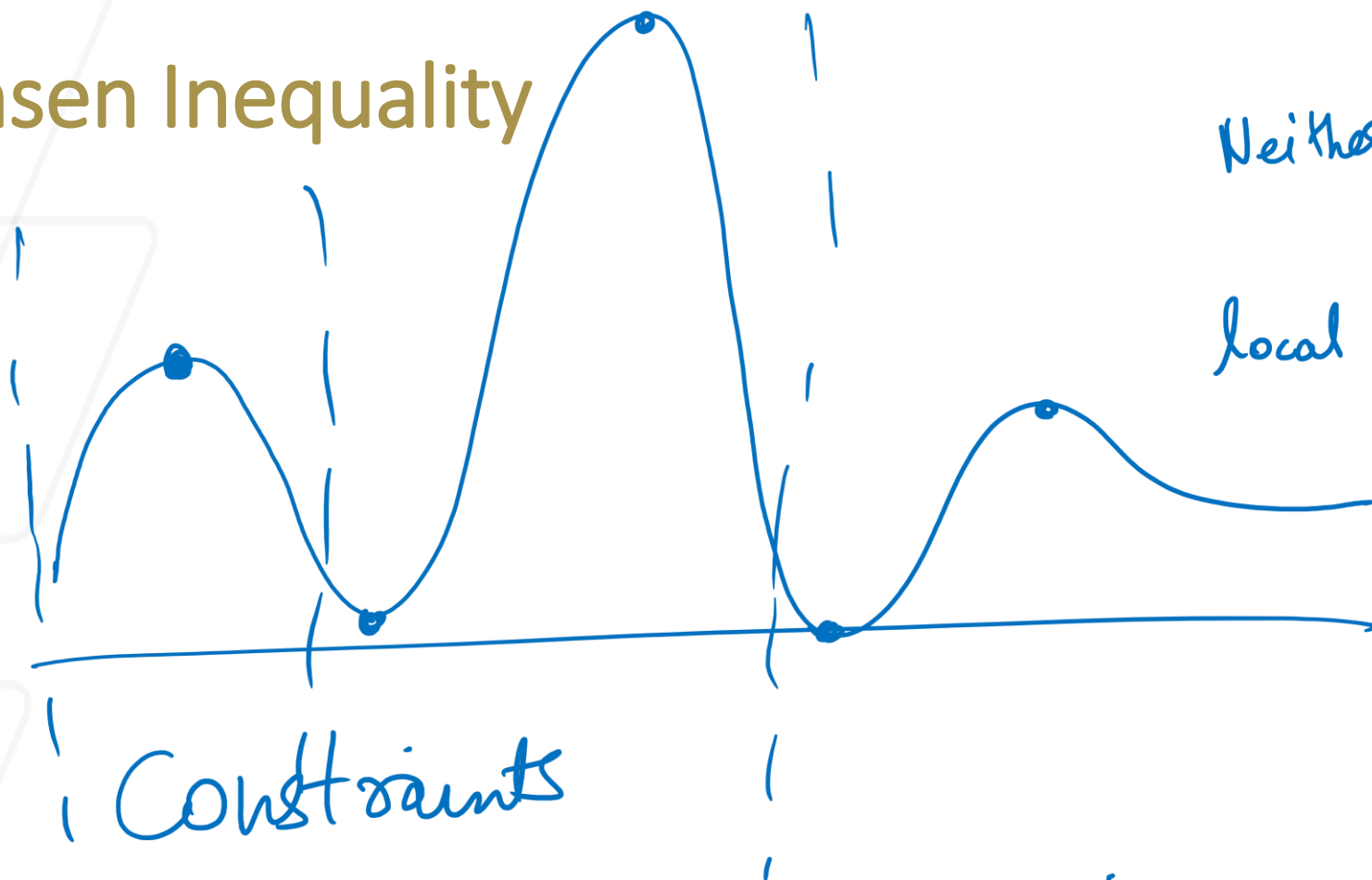
$$f(\mathbb{E}[x]) \quad \mathbb{E}[x] = \sum_x x \cdot p(x) = 1 \times \frac{1}{2} + 2 \times \frac{1}{2} = 1.5$$

Strictly convex functions have global minimum

Jensen Inequality

Neither concave & convex

local maxima & minima



$$\log(x) =$$
$$f'(x) = 1/x$$
$$f''(x) = -1/x^2$$



Concave.

$$E[f(x)] \leq f(E(x))$$

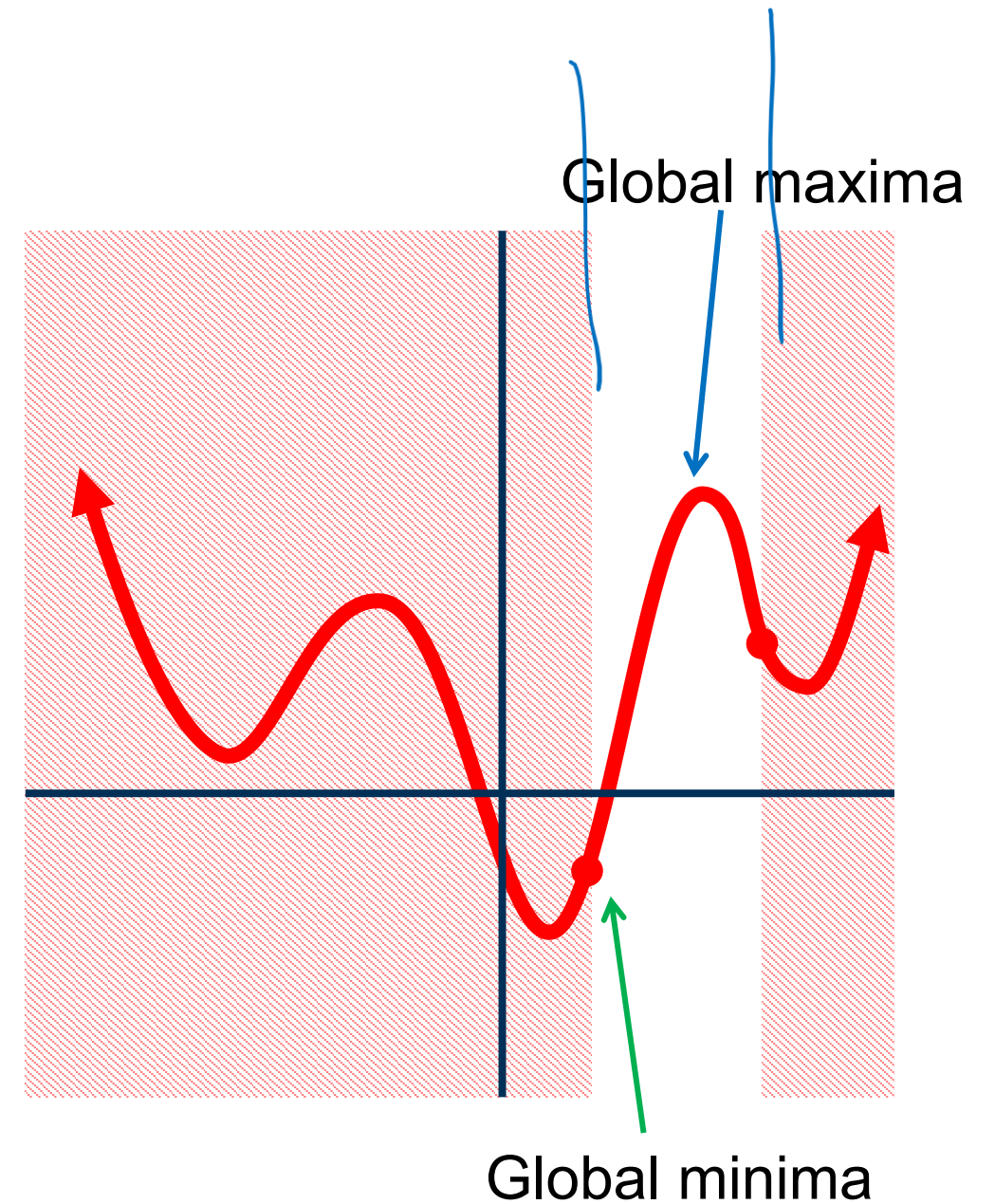
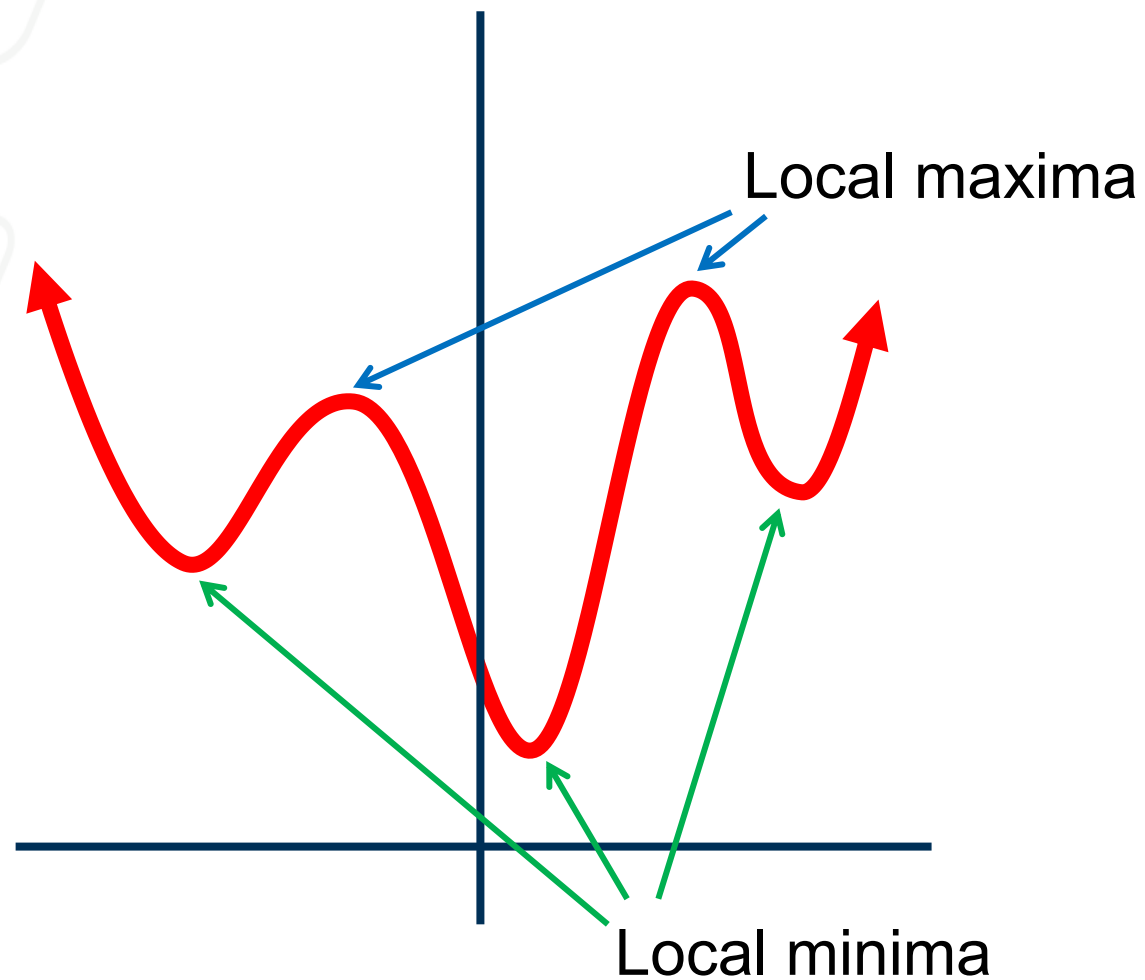
$$E[\log x] \leq \log(E(x))$$

KL Divergence is always non negative

$$\begin{aligned}
 -KL[P][Q] &= \sum_x p(x) \log \frac{q(x)}{p(x)} \\
 &= \sum_x p(x) g(x) \\
 &= E[g(x)] \\
 &= \log(E[q(x)]) \\
 &\leq \log\left(\sum_x p(x) q(x)\right) \\
 &\leq \log\left(\sum_x p(x) \cdot \frac{q(x)}{p(x)}\right) \\
 &\leq \log \sum_x q(x) \leftarrow \text{pdf} \\
 &\leq \log 1 \\
 &= 0
 \end{aligned}$$

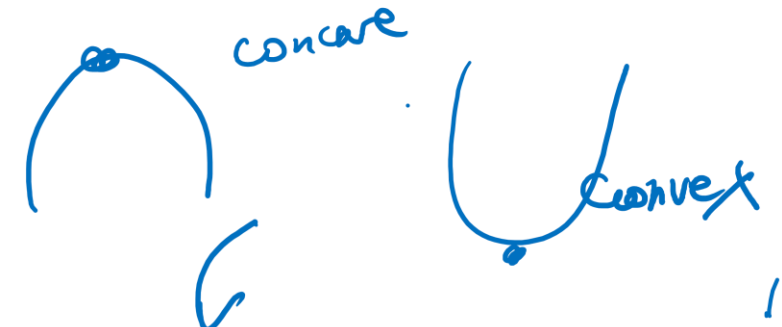
$-KL[P][Q]$
 $KL[P][Q] \geq 0$

Unconstrained and constrained optimization



$$f(x,y) = 6x^2 + 5y^2 \quad \text{Hessian} \begin{bmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial x \partial y} \\ \frac{\partial^2 f(x,y)}{\partial x \partial y} & \frac{\partial^2 f(x,y)}{\partial y^2} \end{bmatrix} \quad \det(\text{Hessian}) < 0 \rightarrow \text{Concave} \\ > 0 \rightarrow \text{Convex}$$

Basic unconstrained optimization problem



- Objective $f(\mathbf{x})$ the quantity we are trying to optimize (maximize or minimize)
- The variables $x_1, x_2, \dots, x_m$ which can be represented in vector form as $\mathbf{x}$
(Note: the subscript m indicates a feature NOT a datapoint in our dataset)
- Finding the stationary points where the function is no longer increasing or decreasing

Equality constrained optimization

$$\min f(x, y) = \frac{6x^2 + 5y^2}{\text{Convex}}$$

→ s.t. $x + y = 1$

- An equality constrained optimization problem is expressed as:

$$\begin{aligned} & \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) \quad \text{objective} \\ & \text{subject to } \underline{g(\mathbf{x}, \mathbf{y}) = 0} \quad \text{s.t.} \end{aligned}$$
$$\underline{g(x, y) = x + y - 1 = 0}$$

Inequality constrained optimization

- An inequality constrained optimization problem is expressed as:

$$\begin{aligned} & \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) \\ & \text{subject to } \underline{g(\mathbf{x}, \mathbf{y}) \leq 0} \end{aligned}$$

K.K.T. conditions

Linear, Quadratic, and Nonlinear Programming

Linear Programming (LP)

Objective: Linear function of x .

Constraints: Linear.

Example: $\min 3x_1 + 5x_2 \text{ s.t. } x_1 + 2x_2 \leq 6$

Quadratic Programming (QP)

Objective: Quadratic

Constraints: Linear.

Example: $\min x^2 + y^2 + 3x + 5y \text{ s.t. } \underline{x + y \leq 4}$

Nonlinear Programming (NLP)

Objective: Nonlinear.

Constraints: Can be nonlinear.

Example: $\min x^2 + y^2 + 3x + 5y \text{ s.t. } x^2 + y \leq 4$



Equality constrained optimization problem

There is a notion of a Lagrangian expression where we combine both the objective function and our constraints

Lagrange $L(\mathbf{x}, \mu) = f(\mathbf{x}) - \lambda g(\mathbf{x})$
s.t. $\lambda \neq 0$

Equality constrained optimization problem

- With multiple equality constraints the Lagrangian is expressed as follows:

$$L(\mathbf{x}, \mu) = f(\mathbf{x}) - \sum_{i=1}^k \lambda_i g_i(\mathbf{x})$$
$$s. t. \quad \lambda_i \neq 0$$

Equality constrained optimization problem

- To solve this optimization problem, we only need to satisfy the stationarity condition, which states:

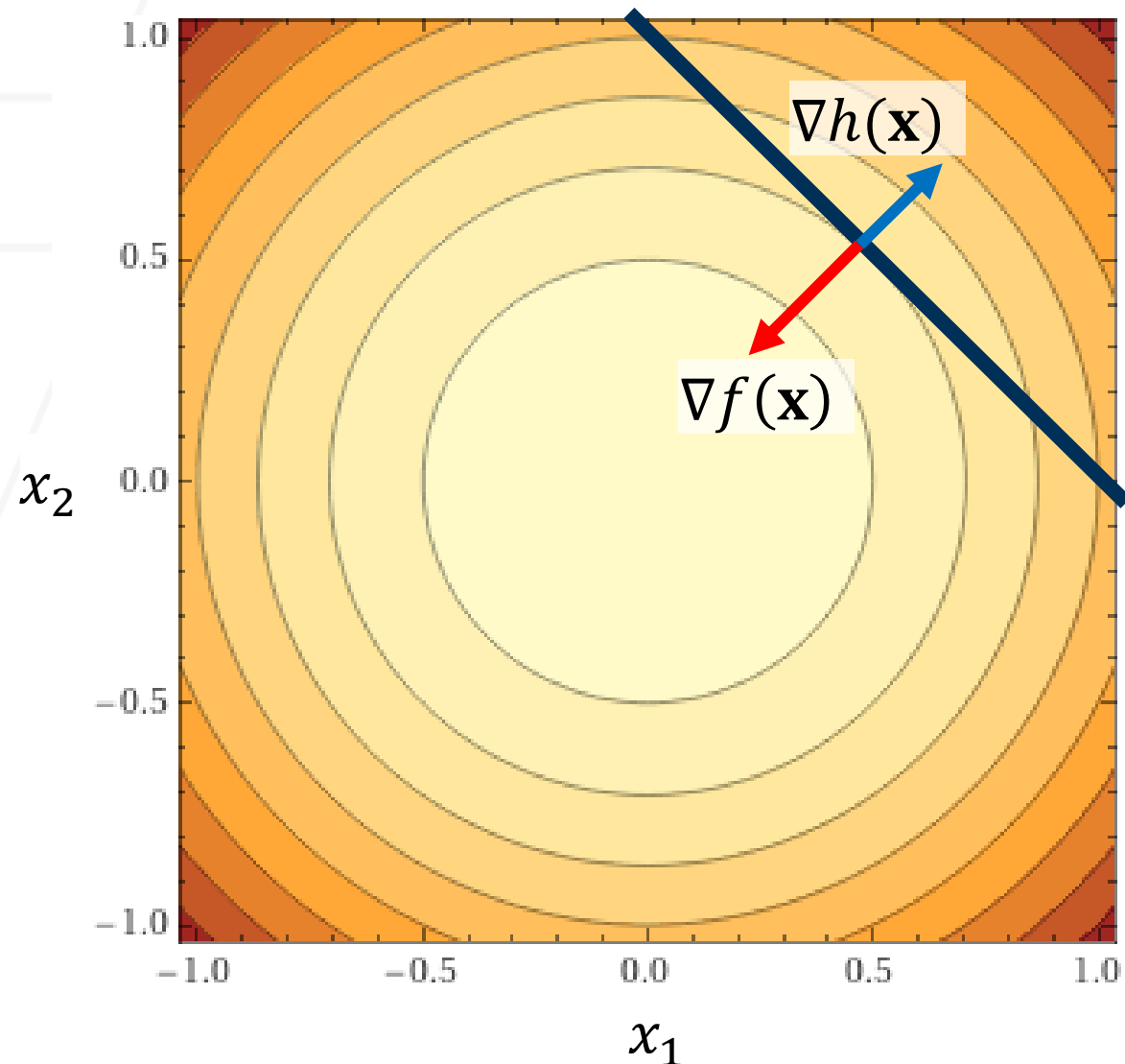
$$\nabla L(\mathbf{x}, \lambda) = 0$$

- We also need to satisfy the primal feasibility condition

$$\lambda_i \neq 0$$

- This appeals to the intuition that at the optimal point, the gradient of the equality constraint is proportional to the gradient of the objective function

Equality constrained optimization: *Example 1*



$$\begin{aligned} \max_{\mathbf{x}} \quad & 1 - x_1^2 - x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1 \end{aligned}$$

Equality constrained optimization: *Example*

$$\begin{array}{ll}\max_{\mathbf{x}} & 1 - x_1^2 - x_2^2 \\ \text{s.t.} & x_1 + x_2 = 1\end{array}$$

1. Write the Lagrangian for this problem

$$L(\mathbf{x}, \lambda) = 1 - x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$$

2. Calculate the gradient with respect to $\mathbf{x}$

$$\frac{\partial L}{\partial x_1} = -2x_1 + \lambda = 0 \rightarrow \lambda = 2x_1$$

$$\frac{\partial L}{\partial x_2} = -2x_2 + \lambda = 0 \rightarrow \lambda = 2x_2$$

Equality constrained optimization: *Example*

3. Solve for $\mathbf{x}^*$ and λ

Using our previous two derivatives and the equality constraint, we obtain:

$$x_1 + x_2 = 1 \rightarrow x_1 + x_1 = 1 \rightarrow x_1^* = x_2^* = \frac{1}{2}$$

From $\frac{\partial L}{\partial x_1} = 0$ we find the value of λ

$$\lambda = 2x_1 \rightarrow \lambda = 1$$

Inequality constrained optimization problem

- Constraints that limit how small or big variables can be. These can be equality constraints, noted as $h_i(\mathbf{x})$ and inequality constraints noted as $g_j(\mathbf{x})$
- An optimization problem is usually expressed as:

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{subject to} & g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m \\ & h_i(\mathbf{x}) = 0, i = 1, \dots, k \end{array}$$

- This form is also referred to as the **primal problem**

Inequality constrained optimization problem

- The primal optimization problem can be rewritten in its **dual form** as

$$\max_{\boldsymbol{\mu}, \boldsymbol{\lambda}} \min_{\mathbf{x}} f(\mathbf{x}) + \sum_{i=1}^r \mu_i h_i(\mathbf{x}) + \sum_{j=1}^m \lambda_j g_j(\mathbf{x})$$

subject to $\lambda \geq 0$

- Satisfying the Karush-Kuhn-Tucker conditions on $\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}$ is equivalent to solving both the primal problem and the dual problem simultaneously
- Furthermore, satisfying the KKT conditions is **necessary and sufficient** for optimality

Karush-Kuhn-Tucker Conditions

- Stationarity condition:

$$0 \in \partial \left(f(\mathbf{x}) + \sum_{i=1}^r \mu_i h_i(\mathbf{x}) + \sum_{j=1}^m \lambda_j g_j(\mathbf{x}) \right)$$

- Complementary slackness

$$\lambda_j g_j(\mathbf{x}) = 0, j = 1, \dots, m$$

- Primal feasibility

$$\begin{aligned} g_j(\mathbf{x}) &\leq 0, j = 1, 2, \dots, m \\ h_i(\mathbf{x}) &= 0, i = 1, \dots, k \end{aligned}$$

- Dual Feasibility

$$\lambda_j \geq 0$$

Karush-Kuhn-Tucker Interpretations

- The complementary slackness condition applies only to inequality constraints
- It states that for a given inequality constraint...
$$\dots \text{either } g_i(x) = 0 \text{ or } \lambda_i = 0$$
- Whenever the dual variable $\lambda_i > 0$ and therefore $g_i(x) = 0$, we say that the **constraint is active** or that the **constraint is tight at $\mathbf{x}$**
- The reverse $\lambda_i = 0$ and therefore $g_i(x) < 0$ means that the constraint is inactive, meaning that it is not effectively impacting the solution

Rule: For each inequality constraint, **either** it is active and has a positive multiplier, or it is inactive and its multiplier is zero.

Primal vs Dual Form

Primal Problem

Original optimization problem:

$$\min_{\mathbf{x}} f(\mathbf{x}) \text{ s.t. } g_j(\mathbf{x}) \leq 0, h_i(\mathbf{x}) = 0$$

Dual Problem

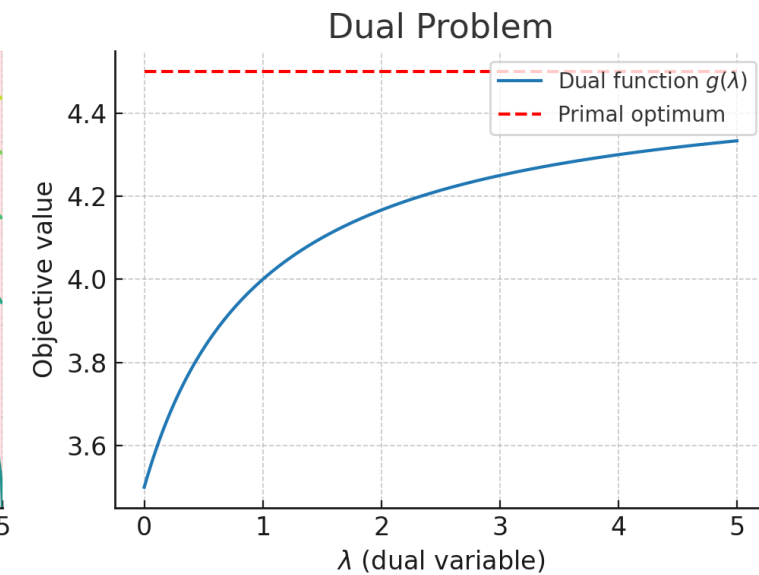
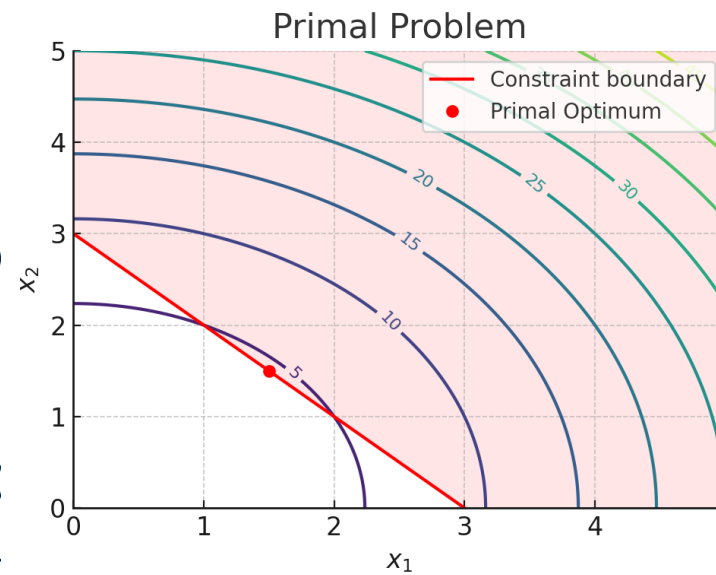
Derived from the **Lagrangian**:

$$L(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \sum_i \mu_i h_i(\mathbf{x}) + \sum_j \lambda_j g_j(\mathbf{x})$$

Dual form:

$$\max_{\lambda \geq 0, \mu} \min_{\mathbf{x}} L(\mathbf{x}, \lambda, \mu)$$

• **Dual variables** λ, μ represent constraint “prices” or sensitivities.



Weak Duality

- Dual optimum $\leq$ primal optimum (for minimization problems).
- Always true.

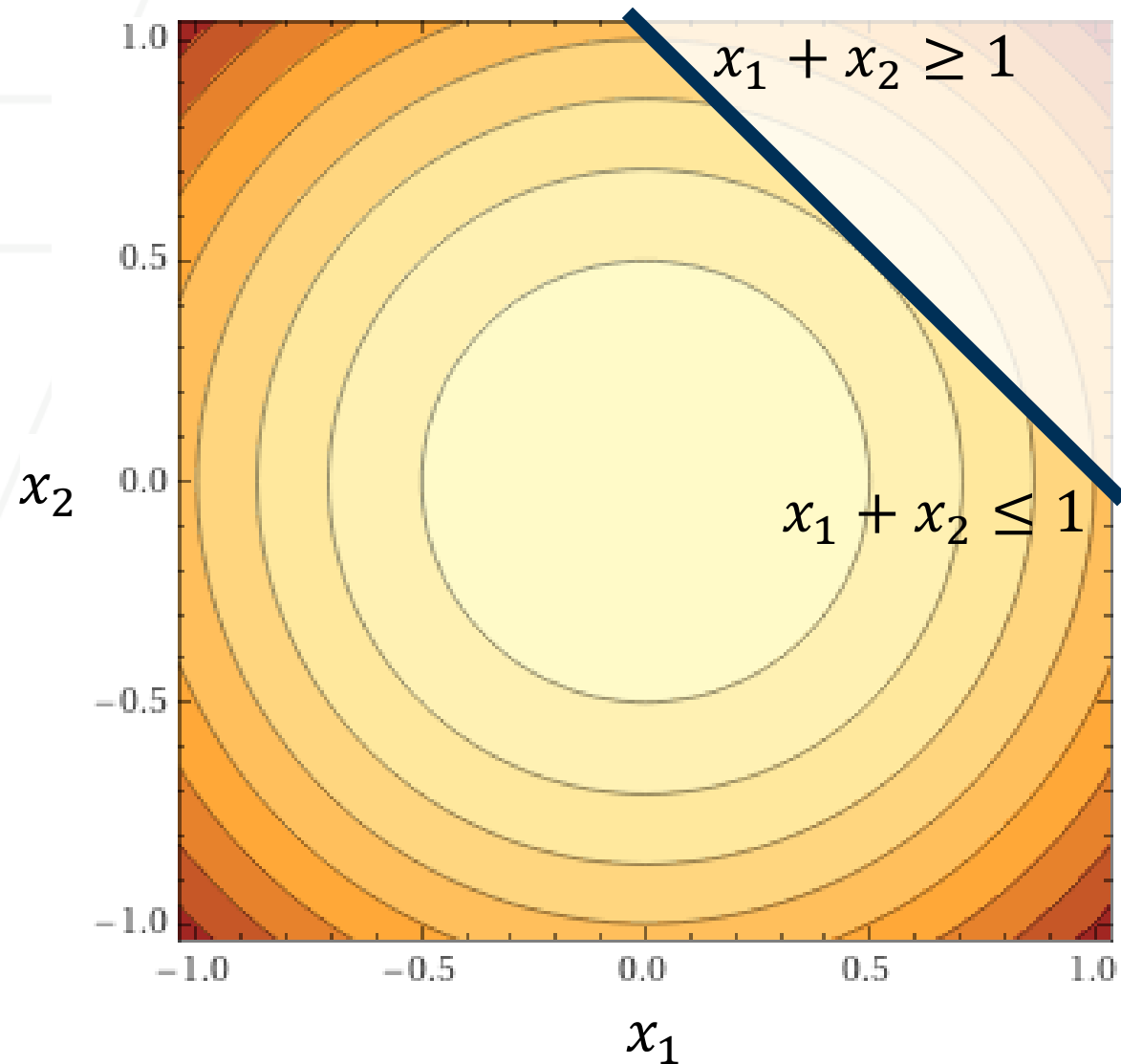
Strong Duality

- Dual optimum = primal optimum.
- Holds for **convex problems** under **Slater's condition** (feasibility + regularity).

Why care?

- Sometimes easier to solve the dual.
- Dual variables give sensitivity info for constraints.
- Many ML models (e.g., SVMs) are solved via the dual.

Inequality constrained optimization: *Example 1*



$$\begin{aligned} \max_{\mathbf{x}} \quad & 1 - x_1^2 - x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 1 \end{aligned}$$

Inequality constrained optimization: *Example 1*

1. Write the Lagrangian for this problem

$$L(\mathbf{x}, \lambda) = -1 + x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$$

2. Calculate the gradient with respect to $\mathbf{x}$ (stationarity condition)

$$\frac{\partial L}{\partial x_1} = 2x_1 + \lambda = 0 \rightarrow \lambda = -2x_1$$

$$\frac{\partial L}{\partial x_2} = 2x_2 + \lambda = 0 \rightarrow \lambda = -2x_2$$

$$x_1 = x_2$$

Inequality constrained optimization: *Example 1*

3. Use the complementary slackness

Case 1: $\lambda > 0$ and $g(x)$
 $= 0$

$$x_1 + x_2 - 1 = 0 \rightarrow 2x_1 = 1$$

$$x_1 = x_2 = \frac{1}{2}$$

4a. Check if the dual feasibility condition is satisfied

$$\lambda = -2x_1 = -1$$

This solution **does NOT satisfy the dual feasibility condition** and it is thus **unfeasible**

Inequality constrained optimization: *Example 1*

3. Use the complementary slackness

Case 2: $\lambda = 0$ and
 $g(x) < 0$

$$\lambda = -2x_1$$

$$x_1 = x_2 = 0$$

4b. Check if the primal feasibility condition is satisfied:

$$g(x_1, x_2) = 0 + 0 - 1 = -1 < 0$$

The solution is feasible and therefore the optimal!

Inequality constrained optimization: *Example 1*

Summary of the KKT conditions for the problem:

$$\lambda = -2x_1$$

$$\lambda = -2x_2$$

$$x_1 + x_2 - 1 \leq 0$$

$$\lambda \geq 0$$

$$\lambda(x_1 + x_2 - 1) = 0$$

Stationarity

Primal feasibility

Dual feasibility

Complementary slackness